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Auditory sensitivity of northern fur seals (*Callorhinus ursinus*) and a California sea lion (*Zalophus californianus*) to airborne sound

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obtained at each primary-signal presentation level. For normal hearing subjects, mean 50% recognition was observed at S/N's ranging from -9 (60 dB SPL) to -5 dB (96 dB SPL) for spondaic words and from -4 (60 dB SPL) to +1.5 dB (96 dB SPL) for monosyllabic words. Performance varied widely among the hearing-impaired. Magnitude of hearing loss, configuration of hearing loss and word-recognition in quiet did not provide a straightforward prediction of performance under difficult communicative conditions. Measurement methods such as those utilized in this experiment may ultimately provide a more accurate prediction of those individuals whose hearing loss severely effects their ability to accurately recognize speech signals under conditions of competition. [Work supported by NINCDS.]

10:30

C9. Paired comparison judgements of relative speech intelligibility. Gerald Studebaker, Jan D. Bisset, and Debra M. Van Ort (Memphis Speech and Hearing Center, 807 Jefferson, Memphis, TN 38105)

The reliability, validity, and sensitivity of paired comparison judgements of speech intelligibility, organized into round robin and elimination tournaments, were studied. Normal hearing and hearing impaired subjects listened to paired samples of speech that were processed with two levels of background noise by 23 hearing aids in two replications. Monosyllabic word intelligibility scores were obtained under each condition in both replications. The results indicated that this paired comparison procedure was a valid, reliable and reasonably sensitive means of evaluating the relative intelligibility of hearing aid processed signals. For example, subjects chose the aid with the best, or second best average discrimination score in 94% of the elimination tournaments. Regarding reliability, the aid that was chosen first on the first trial was the first or second choice on the second trial more than 90% of the time at the 0 dB S/N ratio. The sensitivity of the procedure was such that at the 0 dB S/N ratio, the better aid was chosen in 75% of the matches when the difference between the "true" discrimination scores was in the 9%-12% range. [Work supported by NINCDS.]

10:45

C10. Risk assessment for noise exposure. Linda S. Erdreich (Epidemiological Consultant, 1011 Brookside Drive, Norman, OK 73069) and John Erdreich (University of Oklahoma Health Sciences Center, Department of Otorhinolaryngology, Oklahoma City, OK 73190)

In this paper we propose that full use is not made of available noise-exposed threshold data. Specifically, data from noise-exposed populations can be analyzed to assess impact of exposure in two ways: (1) to provide estimates of the number of cases of hearing loss attributable to noise in the population or, (2) to provide estimates of increased risk of hearing loss in an exposed individual relative to the unexposed person. Previous studies have neglected prediction of risk to the individual, emphasizing instead the impact on the population. To more fully understand the effect of noise exposure, the approaches must be combined. We will present two statistics which may be calculated from population threshold data: *Percent risk* is the statistic commonly computed from hearing conservation data. Percent risk is an estimate of the number or of the proportion of cases of hearing loss attributable to noise exposure. *Rate ratio* is the second statistic we will discuss. The rate ratio estimates the increased risk of hearing loss to an individual as a consequence of exposure. It is adjusted for the background rate and is, therefore, comparable across studies. Each statistic complements the information provided by the other. We will also discuss how potential risk factors for hearing loss may be evaluated and combined to yield a composite risk estimate.

11:00

C11. Auditory sensitivity of northern fur seals (*Callorhinus ursinus*) and a California sea lion (*Zalophus californianus*) to airborne sound. R. J. Schusterman (Departments of Psychology and Biology, California State University, Hayward, CA 94542) and P. W. B. Moore (Naval Ocean Systems Center, P.O. Box 997, Kailua, HI 96734)

Aerial audiograms were obtained in a specially constructed acoustic chamber from three otariid pinnipeds—two yearling female *Callorhinus* and one 2-yr-old female *Zalophus*. A "go/no-go" response procedure was used and threshold estimates were obtained by a tracking method. Behavioral measurements at 1, 2, 4, 8, 16, 24, and 32 kHz resulted in average thresholds for the two *Callorhinus*, in the order of frequencies given above, of 29, 9, 22, 13, 7, 22, and 34 dB re 0.0002 dyn/cm², and for the one *Zalophus* they were, respectively, 41, 19, 26, 16, 28, 37, and 61 dB re 0.0002 dyn/cm². The thresholds for *Callorhinus*, although inferior in air compared to in water, show good accommodation for hearing airborne sounds. The present results from *Zalophus* indicate that a previous study [R. J. Schusterman, J. Acoust. Soc. Am. 56, 1248-1251 (1974)] was noise limited in threshold determinations below 24 kHz. Although only a limited number of species have been looked at, the otariid pinnipeds appear to be more sensitive to airborne sounds than do the phocid pinnipeds. [Work supported by ONR.]